How to run system with one command line (Requires CMAKE to be installed):

Commands from root:
> cd Frontend
> ./build.bat
> build/debug/frontend.exe current_user_accounts.txt available_games.txt game_collection.txt daily_transaction.txt

The final line above is one command line, and can be typed in terminal from the frontend folder to use the program

An alternative would be to use run.bat using ./run.bat

Alternative:

Open terminal in the frontend directory (/Frontend)
Run : g++ -Iinclude src/*.cpp -o frontend
Then run:./frontend.exe current_user_accounts.txt available_games.txt game_collection.txt daily_transaction.txt
this could be run as one command line from the frontend folder (simpler command line)

How to run tests:

From root, run run_tests.bat
Can be done in terminal using: ./run_tests.bat

A folder named "test_runs" will be made and contain each test case's actual outputs, as well as test summaries to see which test cases passed and failed.

Run 1: **ALL TESTS FAIL**

#	Test Cases	Tests	Reason	Fix
——	**Authentication**	x4 Tests		
T01	loginExistingAccount	Success: **FAILURE** Failure: **FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated Login::startSession(), Login::login(), and terminal formatting to display the expected login results.
T02	loggedoutTransactions	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated runtime::loggedIn() added 0.00 to the input Corrected the expected username to RegularUser20.
T03	loggedIn	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated Login::login() and corrected the expected prompt sequence.

T04	logoutWhileLoggedIn	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated main() to report the specific logout error
——	**Account Creation**	x10 Tests		
T05	maxUsernameChar	Success: **FAILURE** Failure:**FAILURE**	Success Failed due to missing input for amount of money	Added early validation through checkCreateName() in main() added 0.00 for the valid account.
T06	uniqueUsername	Success: **FAILURE** Failure:**FAILURE**	Success Failed due to missing input for amount of money	Updated checkCreateName() to reject duplicates immediately Added the missing credit input.
T07	createFS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated the creation handler in main() to display the expected permission error.
T08	createBS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated the creation handler in main() to display the BS-specific error.
T09	createSS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated the creation handler in main() to display the SS-specific error.
T10	adminCreateAA	Success: **FAILURE** Failure:**FAILURE**	Success Failed due to missing input for amount of money	Added 0.00 to the input and aligned the creation prompts and success message in main().
T11	adminCreateBS	Success: **FAILURE** Failure:**FAILURE**	Success Failed due to missing input for amount of money	Added 0.00 and aligned creation output in main().
T12	adminCreateFS	Success: **FAILURE** Failure:**FAILURE**	Success Failed due to missing input for amount of money	Added 0.00 and aligned creation output in main().
T13	adminCreateSS	Success: **FAILURE** Failure:**FAILURE**	Success Failed due to missing input for amount of money	Added 0.00 and aligned creation output in main().

T14	createNonExistingType	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated main() to validate the account type before requesting credit.
——	**Account Deletion**	x7 Tests		
T15	deletePermsAA	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated deletion messages in main() and corrected the expected prompt.
T16	deletePermsFS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated main() and added the expected FS logout transaction file.
T17	deletePermsBS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated the deletion permission message in main().
T18	deletePermsSS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated main() and replaced the placeholder with the five game test catalog.
T19	deleteCurrUser	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated the error in DeleteUser::removeUser().
T20	deleteExistingUser	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated main() to check existence and display the expected deletion error.
T21	transactionsOnDeletedUser	Success: **FAILURE** Failure:**FAILURE**	Success Failed due to missing input for amount of money	Added 10.00 before the username updated main() and AddCredit::updateCredit() to report the deleted account correctly.
——	**Buying Games**	x7 Tests		
T22	buyExistingGame	Success: **FAILURE** Failure: **FAILURE**	Success Failed due to missing input for amount of money	Created a five-game test baseline and added early game validation in main().

T23	buyAA	Success: **FAILURE** Failure: **FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated purchase output in main() and aligned catalog formatting.
T24	buyFS	Success: **FAILURE** Failure: **FAILURE**	Success failed due to code error	Corrected the isolated test baseline. Retained ownership protection in Buy::execute() FS buying permission already worked.
T25	buyBS	Success: **FAILURE** Failure: **FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated purchase prompts and confirmation in main().
T26	buySS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated the buy permission error in main().
T27	withinCreditLimit	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated the insufficient credit message in Buy::execute().
T28	alreadyOwned	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated ownership errors in Buy::execute() and Buy::addGame().
——	**Selling Games**	X8 Tests		
T29	sellAA	Success: **FAILURE** Failure:**FAILURE**	Success failed due to code error	Used the five game test baseline Aligned sale validation and confirmation in main() and Sell::execute().
T30	sellFS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to code error	Corrected the test baseline and aligned sales output. FS permission remained unchanged.
T31	sellSS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to code error	Corrected the test baseline and aligned sales output. SS permission remained unchanged.

T32	sellBS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated the selling permission error in main().
T33	maxPrice	Success: **FAILURE** Failure:**FAILURE**	Success failed due to code error	Corrected the baseline added explicit price validation in Sell::execute() Corrected 050.0 to 050.00.
T34	gameNameLen	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Added early title validation through checkSellTitle() in main().
T35	uniqueGameName	Success: **FAILURE** Failure:**FAILURE**	Success Failed due to missing input for amount of money	Added early title validation through checkSellTitle() in main().
T36	newGameTransaction	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Added same session checks in main() and Buy::execute() corrected the test baseline.
——	**Refunds**	x9 Tests		
T37	refundAA	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Added same session checks in main() and Buy::execute() corrected the test baseline.
T38	refundSS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated the refund permission error in main().
T39	refundFS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated the refund permission error in main().
T40	refundBS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated the refund permission error in main().
T41	checkRefundCred	Success: **FAILURE** Failure: **FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated the seller credit error in Refund::execute() and corrected the expected refund to $65.00.

T42	checkMaxCredRefund	Success: **FAILURE** Failure: **FAILURE**	Success failed due to code error	Fixed input order in main() and corrected the maximum balance constant. Retained the buyer limit check in Refund::execute().
T43	checkBuyerRefundAmt	Success: **FAILURE** Failure: **FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated refund confirmation in main() and the transfer logic in Refund::execute().
T44	refundNoOwn	Success: **FAILURE** Failure: **FAILURE**	Success Failed due to missing input for amount of money	Added an early ownership check in main() and seller related ownership validation in Refund::execute().
T45	existingAccountRefund	Success: **FAILURE** Failure: **FAILURE**	Success Failed due to missing input for amount of money	Added explicit account validation in Refund::execute() corrected expectations to reject the missing seller and show Game 3 for the valid refund.
——	**Add Credit**	x7 Tests		
T46	creditAA	Success: **FAILURE** Failure:**FAILURE**	Success failed due to code error	Changed the add credit handler in main() to read the amount before the recipient.
T47	creditBS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated add-credit output in main().
T48	creditFS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated output in main() and corrected the expected logout type to FS.
T49	creditSS	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated add credit and confirmation in main().
T50	existingAccountCredit	Success: **FAILURE** Failure:**FAILURE**	Success failed due to code error	Fixed input order in main() and added the explicit missing recipient error in AddCredit::updateCredit().

T51	maxAdd	Success: **FAILURE** Failure:**FAILURE**	Success failed due to code error	Updated AddCredit::updateCredit() to check the session total before changing balances. Corrected input
T52	maxCred	Success: **FAILURE** Failure:**FAILURE**	Success failed due to incorrect output on terminal, code functionality works as intended	Updated main(), corrected MAX_CREDIT to 99999999 cents, and aligned the balance limit message in AddCredit::updateCredit().

Run 2: **Failure Tests Fail**

#	Test Cases	Tests	Reason	Fix
T01-T52		Success: **PASS** Failure: **FAILURE**	Expected outputs for failure tests accept invalid inputs	Made failure cases expected outputs reject invalid inputs

Run 3: **Failure Tests Pass**

#	Test Cases	Tests	Reason	Fix
T01-T52		Success: **PASS** Failure: **PASS**	Expected outputs for failure tests accept invalid inputs	N/a All test run correctly
——	**Listing Accounts**	x2 Tests		
T53	listAccounts	Success: **PASS** Failure: **PASS**	Expected outputs for failure tests accept invalid inputs	N/a All test run correctly
T54	listAccountsPerms	Success: **PASS** Failure: **PASS**	Expected outputs for failure tests accept invalid inputs	N/a All test run correctly
——	**Seach**	x3 Tests		
T55	searchResults	Success: **PASS** Failure: **PASS**	Expected outputs for failure tests accept invalid inputs	N/a All test run correctly
T56	searchNonAdmin	Success: **PASS** Failure: **PASS**	Expected outputs for failure tests accept invalid inputs	N/a All test run correctly
T57	searchAdmin	Success: **PASS** Failure: **PASS**	Expected outputs for failure tests accept invalid inputs	N/a All test run correctly

New Test Cases added

#	Test Cases	Intended Test
——	**Listing Accounts**	x2 Tests
T53	listAccounts	Check if the the listaccounts transaction lists all existing accounts
T54	listAccountsPerms	Checks if only admin can use the listaccounts transaction
——	**Seach**	x3 Tests
T55	searchResults	Check if search results are only results that contain the the search query
T56	searchNonAdmin	Check if admin users who aren't selling games aren't shown if the user who searched isn't an admin
T57	searchAdmin	Check if admin users who aren't selling games are shown if the user who searched is an admin

Updated Files:

main.cpp
Runtime.inc
Login.cpp
GameCatalogPage.cpp
Buy.cpp
Sell.cpp
Refund.cpp
AddCredit.cpp
DeleteUser.cpp